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Minimizing Flaring Through Start-Up and Shutdown Optimization: A Multi-Console Operational Success

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Abstract

This study presents a successful flare reduction initiative achieved by optimizing start-up and shutdown (SD) sequences across a critical gas processing unit. The effort focused on enhancing coordination between four operation consoles: Utility, Liquid Handling, Gas Processing, and Gas Compression. By improving real-time communication and aligning critical steps, the team significantly reduced flaring volumes, supported environmental goals, and enhanced operational discipline during transient conditions.

A cross-functional task force was formed to review historical flaring data, SD/start-up procedures, and console roles. Workshops and simulations were held to map interdependencies and identify misalignments in communication timing between Utility, Liquid Handling, Gas Processing, and Gas Compression consoles. An optimized stepwise procedure was developed and embedded with clear handshakes and communication triggers between consoles. Operators were trained using real-case simulations, emphasizing discipline in following the new sequences. Shift alignment, start-up readiness checklists, and console-specific accountability were also implemented to ensure consistent execution across different shifts and operating scenarios in pressure stabilization, avoiding unnecessary venting. The Utility console's timely readiness checks minimized delays, allowing early steam and air integration, which also helped prevent pressure surges and excess purge gas flaring.

This study demonstrates how targeted operational improvements not major capital investments can deliver significant flaring reductions. The novelty lies in a practical and replicable approach: optimizing inter-console communication and synchronization across Utility, Liquid Handling, Gas Processing, and Gas Compression consoles. It highlights the often-overlooked human and procedural aspects of flare reduction during transient operations, offering a scalable model for facilities aiming to reduce emissions and enhance control room coordination.

INTRODUCTION

Petroleum Development Oman (PDO) operates in the south-eastern part of the Sultanate of Oman. The South Sour Cluster ([Figure 1](#)) comprises four production stations handling more than 24 reservoirs. Reservoir pressures are high and range between 500-1000 Bar G.

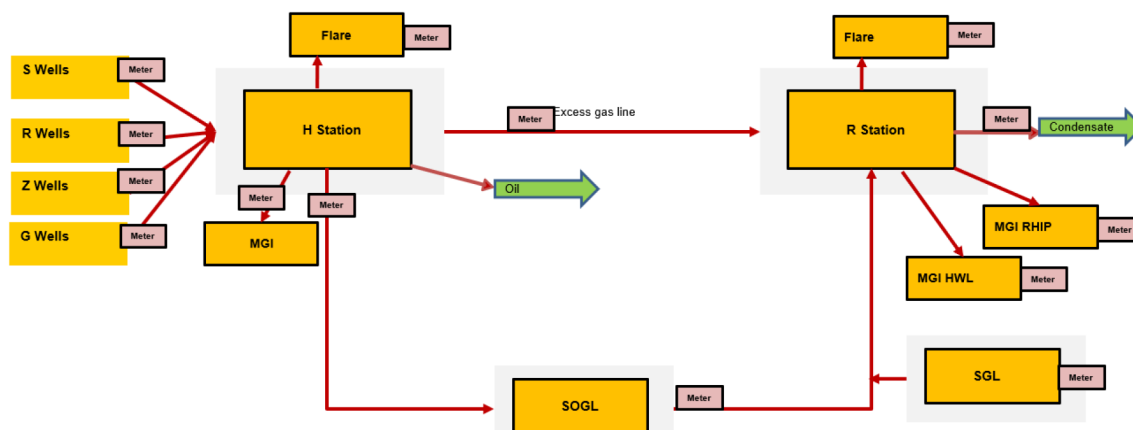


Figure 1—South Oman Sour Cluster Overview

The intent of the Station R is to maximize the liquid recovery from gas/condensate reservoir) by means of partial recycling and from the Harweel oil reservoirs by means of Miscible Gas Injection (MGI), whilst providing sweet gas to the market [figure 1](#).

R plant is operated to produce the following:

- Sweet sales gas, which is transported via the Salalah gas pipeline.
- Dehydrated sour gas, which is re-injected to the H plant oil reservoirs for Miscible
- Gas Injection (MGI)
- Dehydrated sour gas for injection into the R gas reservoir to maintain reservoir
- pressure.
- Sweet Hydrocarbon condensate, which is transported via the Main Oil Line (MOL).
- Wastewater, which is injected into a Deep-Water Disposal (DWD) well.

The R Station consists of four main units:

- Utility
- Liquid Handling.
- Gas Treatment.
- Gas compression.

Utility

Water treatment Plant

The purpose of this plant is to treat raw well water to produce dematerialized water, potable water and utility water to be used within the R station. Water received from wells is chlorinated & then sent to Raw Water Tanks well as the Fire water tanks. Raw Water Transfer Filtered Water Storage Tanks after passing through the Multimedia Filters to have the suspended solids removed, but with coagulant dosing upstream of the Multimedia filters & De-chlorination dosing downstream. The filtered water feed to Reverse Osmosis System is dosed with Anti-Scalant to get pH around 7.4, it is then passed through the RO Cartridge Filters to reduce further the particle size to acceptable limits to the RO system. RO System will reduce the dissolved solids to low level acceptable to the downstream Demineralized Water Mixed Bed System. RO reject will

be routed to Effluent Water Storage Tanks while RO permeate will be fed to Vacuum deaeration system. Vacuum deaeration system will remove the oxygen less than 10 ppb for the downstream units. Dearated water is then pumped via Dearated Water Transfer Pumps to the Pretreated Water Storage Tanks and a small side stream of (10%) maximum to the Potable Water Tank after its potabilized via dosing with carbon dioxide, treatment in Dolomite Filter, sodium hypochlorite dosing and finally polishing in Potable Water Cartridge Filter, from this tank its then distributed to users.

Heat Recovery Steam Generation (HRSG)

During power generation in the power plant (utilizing gas turbine), high temp flue gas at 425-600 Celsius is produced. If this hot exhaust gas is directly released to atmosphere, it is clearly a large loss of thermal energy. If some of this heat loss can be recovered and converted to useful energy, the process efficiency will be increased with both economic and environmental benefits. This is achieved by circulating demineralised water through the path of hot exhaust gas from the gas turbine. Here the demineralised water is converted in to saturated or superheated steam of desired pressure which can be further used for some utility or Industrial purposes. During this process the temperature of the flue gas leaving the stack is lowered. These heat recovery units which are installed at the downstream of the Gas turbines to produce the steam are termed as 'Heat Recovery Steam Generation'.

Other system in Utility:

- HP & AP Flare systems
- Closed Drain system (sour & sweet)
- Open Drain System (hazardous & non-hazardous)
- LP / HP Fuel Gas systems
- Seal gas Compression system
- Fire Water System
- Instrument Air System and N2 system
- Chemical Injection Systems
- Diesel system for Fire water pumps

Liquid Handling

The purpose of the Inlet Separation Facilities is to remove Condensate (HC liquid and Produced Water) from the feed gas. The separated gas is sent to Dehydration unit of MGI compression section for Miscible Gas Injection and partial recycling, Train-1/Train-2 of GSU to produce sweet sales gas for export to SGL. There are three Inlet Separators available to treat the feed gas. H Inlet Separator receives feed gas from H station excess Sour gas pipeline and the separated gas is sent to H MGI inlet Manifold for dehydration, compression and to injection. R separators receive feed gas from R Remote Gas Gathering Manifold are sent to Train-1/Train-2 of GSU to produce sweet sales gas for export to SGL. Excess sour gas is sent to the H MGI inlet Manifold for dehydration, compression and to inject to H and R injection wells. Separated liquid from all three Inlet Separators are sent to the Condensate Stabilization unit through pre-conditioning units and De-Salter. Condensate Stabiliser is to treat the feed HC Condensate received from Condensate Desalting package to meet the specification of condensate for storage and export to the pipeline.

Gas Treatment

The purpose of the Gas Sweetening Unit (GSU) trains ($2 \times 50\%$ train configuration) is to sweeten ~ 8.8 MMSm³/d of peak sour gas containing 4.7mol% H₂S and 21.7 mol% CO₂. A portion of sour gas from the

R and Test Separator is mixed with sour flash gas from the HP Flash Gas Compressor Discharge KOD prior to entering the GSU. The feed gas goes through the inlet knockout drums where it is washed with sour water pumped by Acid Water Flash Vessel Sour Water Pumps, for removal of salts prior to its entry into the GSU Absorbers. Each GSU train is operated based on regenerative solvent Sulfinol $\times$ Process designed to reduce the acid gas components H_2S and CO_2 to meet the sales gas specification of 5 ppmv and 3.5 mol% respectively in combination with COS / Mercaptan removal in the downstream Molecular Sieve Unit (MSU) beds. The solvent is contacted counter-currently with the sour feed gas in absorption columns. The treated gas leaves at the top and eventually flows to the MSU. The gas leaving the GSU is saturated with water and still contains some of the COS / Mercaptan compounds. COS content is reduced to less than 50 ppmv due to deep COS removal requirements meeting total Sulphur specification in downstream MSU. HC Dew Pointing unit is to cool down the dry sweet gas from the MRU unit to attain a Cricondentherm temperature of $14^\circ C$ by the R-134a Refrigerant Compressor to avoid hydrocarbon condensation in the Sales Gas pipeline. Then the export gas compressor increases the pressure to pass through the South Oman Gas Line.

Gas compression

The purpose of R Sour Gas Dehydration Unit is to dehydrate feed Gas received from Inlet Separation Facilities, Molecular Sieve Unit along with 50% acid gas from Acid Gas Compression Package, using Lean TEG, to achieve moisture content specification. On-spec dry gas is sent to R Gas Injection Compression Unit is to boost the pressure of dehydrated gas from Rabab Sour Gas Dehydration Unit from to 36100 kPag and injected to R Gas Injection Wells figure 2.

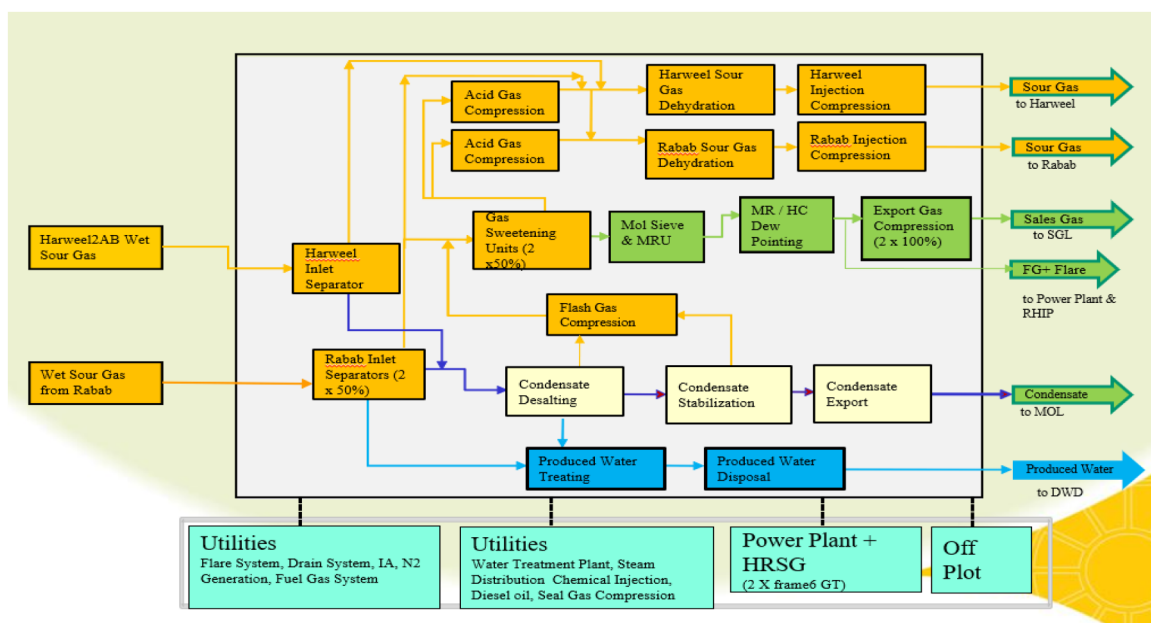


Figure 2—R station Overview

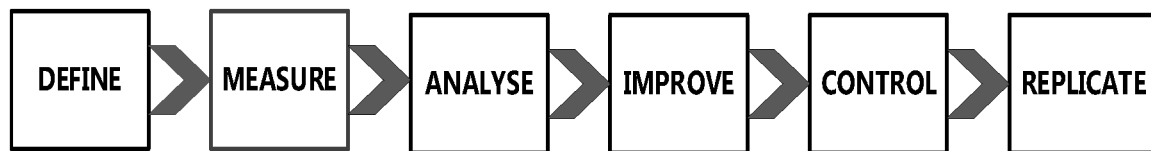
Lean Six Sigma (LSS) APPROACH

LSS is a disciplined, data-driven approach to process improvement that aims to eliminate defects, reduce process variation and continuously improve operation conditions. There are several tools and techniques used in Six Sigma methodology to analyze and improve processes.

The DMAIC approach is employed in LSS projects. This approach encompasses a closed-loop process consisting of sequential steps aimed at enhancing an already existing business process.

The DMAIC project methodology comprises five distinct phases:

1. Defining the focal system, which involves capturing the voice of the customers along with their requirements and expectations, as well as setting specific project goals.
2. Measuring, which entails assessing key elements of the current process and gathering all pertinent data, while also evaluating the capability of the existing process.
3. Analyzing, which delves deeply into the data to explore all potential sources of variation and establish cause-and-effect relationships.
4. Improving or adjusting the current process, which is carried out based on the insights gained from the data analysis.
5. Controlling the future state process, which involves ensuring the sustainability of the improvements by promptly addressing any deviations from the desired target to prevent defects.



Here are some commonly used tools we applied:

- **SMART Goal:** Having a Specific, Measurable, Achievable, Realistic and Time-based value goal has helped to measure the success at the end of the project.
- **PROCESS MAPPING:** Process mapping is about creating a picture of how a process works. It shows the steps, things going in and out, and how everything is connected. By doing this, we can find ways to make the process better. "Operations, Inputs, Process, Outputs, and Implementation" is a tool that helps us understand the important parts of a production unit and the people involved in it. In the DMAIC approach, detailed process mapping is commonly used where all inputs and outputs for each step are mapped. The team then identifies which input (or factor or x) can be controlled. All identified factors (x) then go through graphical and statistical analysis.
- **CAUSE AND EFFECT DIAGRAM (Fishbone diagram):** This tool is used to identify and categorize potential causes of a problem or defect. It helps explore different factors that may contribute to the issue which might be missed from the detailed process map.
- **PARETO CHART:** is a bar chart that ranks problems or causes in descending order of frequency or impact. It helps identify the vital few factors that account for many problems, enabling focus on the most significant issues.
- **GRAPHICAL ANALYSIS:** Where several tools like scatter plot, matrix plot, main effect plot and dot plot are used to filter the key factors affecting the main output.
- **HYPOTHESIS TESTING:** Is used to make data-driven decisions by statistically testing hypotheses in the process equipment or system. It helps determine whether observed differences or relationships are statistically significant.
- **DESIGN OF EXPERIMENTS:** It is a systematic approach used to understand and optimize process variables and their interactions. It allows experimentation to identify the optimal settings for process parameters.
- **FAILURE MODE AND EFFECTS ANALYSIS:** It is a proactive tool used to identify and mitigate potential failures or defects in a process, production, or system. It helps us to assess the severity, occurrence, and detectability of potential failure modes. These are just a few examples of the many

DEFINE Phase. The R plant is characterized by its complex nature and extensive array of equipment. It manages an average daily production of condensate at 8500 cubic meters, dealing with challenging

conditions such as high hydrogen sulfide content (over 6% reaching 60000 ppm) and high pressure (up to 500 bar). Additionally, the plant process natural gas and condensate from the gas wells, along with associated gas production from H station, which transported to R station via a crossover at rate of approximately 6 million standard cubic meters per day. The time required to reactivate wells after shutdown varies, leading so significant flaring of greenhouse gas production deferment. Gas oil ratio (GOR) also contributing to inefficiency, including defects overpressure, overproduction, and delay. Moreover, R is connected to power station, where steam generation plays a crucial role in reducing fuel consumption.

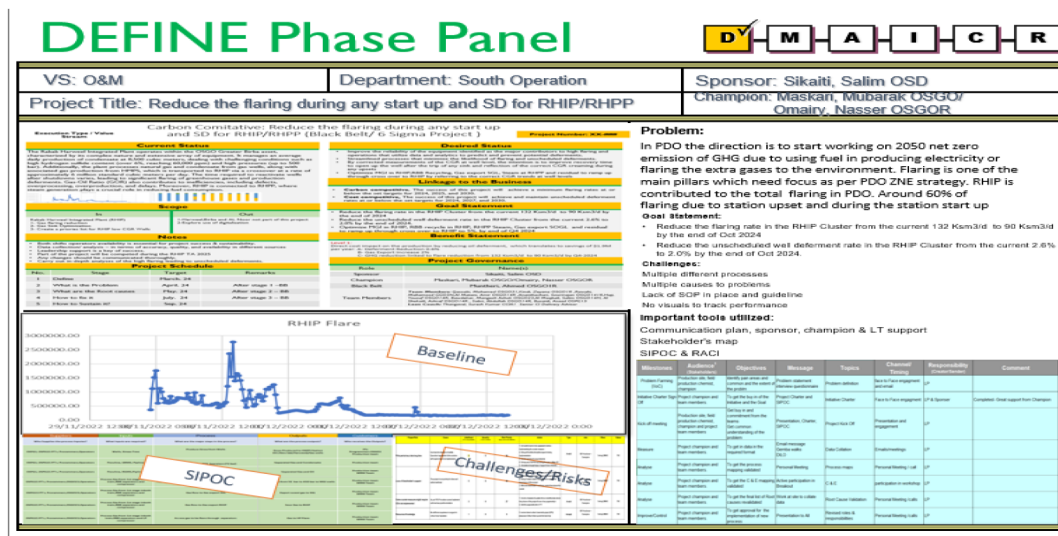


Figure 3—Define Phase

Problem

In PDO the direction is to start working on 2050 net zero emission of GHG due to using fuel in producing electricity or flaring the extra gases to the environment. Flaring is one of the main pillars which need focus as per PDO strategy. R station is contributed to the total flaring in PDO. Around 60% of flaring due to station upset and during the station start up.

The main goal statements

- Reduce the flaring rate in the R Cluster from the current 132 Ksm3/d to 90 Ksm3/d by the end of Oct 2024.

Challenges

- Multiple different processes
- Multiple causes to problems
- Lack of SOP in place and guideline
- No visuals to track performance

Important tools utilized

- Communication plan, sponsor, champion & LT support
- Stakeholder's map

- SIPOC & RACI

MEASURE Phase

It measures the current performance by collecting the data to understanding the baseline.

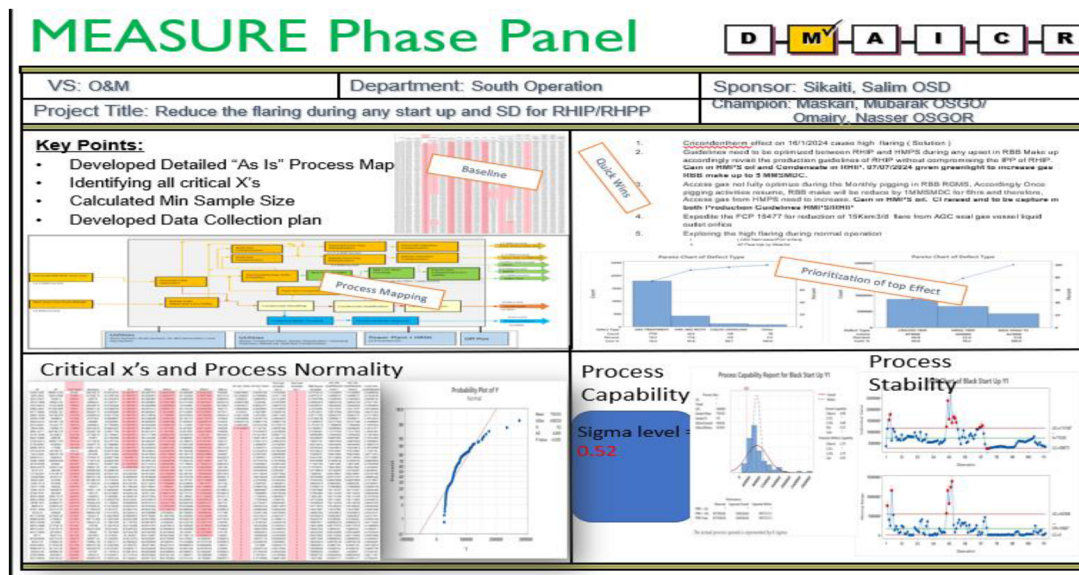


Figure 4—Measure Phase

Key Points:

- Developed Detailed "As Is" Process Map
- Identifying all critical X's
- Calculated Min Sample Size
- Prioritization of top effect
- Developed Data Collection plan

ANALYSE Phase

The primary tools used for data graphical analysis included Scatter Plot, Matrix Plot, Box Plot, Correlation, Regression, Normality Test, and 2-Sample Variance, all of which helped visualize and understand the behavior and relationships within the flaring data. Through these visual tools, it was observed that certain process units showed a higher correlation with increased flaring events. Specifically, regression analysis revealed that the Gas Treatment (GT) unit and the Heat Recovery Steam Generators (HRSG) were the most significant contributors to high flaring levels. Based on these findings, hypothesis testing will be conducted to validate two key assumptions:

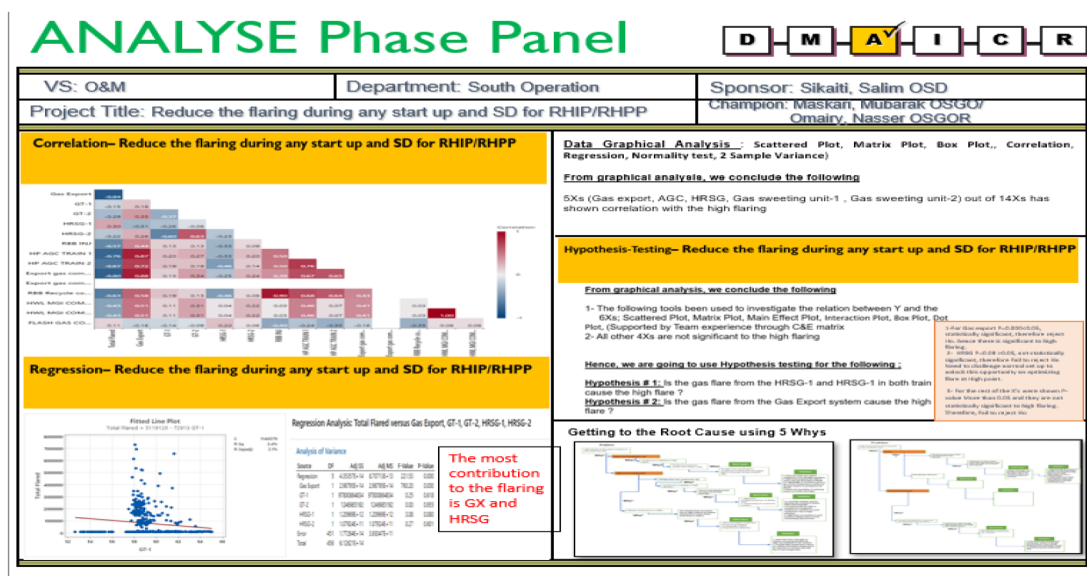


Figure 5—Analyze Phase

Hypothesis 1: the R facility operates with 2×100% HRSGs, though under normal conditions, each HRSG typically runs at 50% load. Approximately 90% of the steam generated is consumed by GSU Trains 1 and 2. When one of the HRSGs trips, the resulting steam shortfall causes the treated gas from the GSUs to go off-spec, leading to flaring until the HRSG is restored. To mitigate continuous flaring during such events, operations often ramp down well production. However, this creates a secondary challenge: the acid gas from both GSU trains, along with lean gas from TEG regeneration, VRC, and other sources, is processed by 2×50% acid gas compressors. Reducing well production decreases the volume of acid gas to the compressors, lowering the molecular weight of the inlet gas and causing the compressors to unload. This in turn increases flaring from GSU regeneration, TEG regeneration, and VRC systems. Furthermore, combustion of the acid gas requires assist gas from the fuel gas system, further raising the total flaring demand. Due to the complexity and environmental impact of this issue, a detailed study was conducted to identify and implement mitigation strategies to reduce flaring under HRSG trip scenarios.

Hypothesis 2: The R facility exports approximately 5–5.5 MMSCMD of sweet gas to downstream facilities. To meet required gas specifications, a dew point control unit is in place, maintaining a cricondentherm temperature of 17°C using 3×50% refrigeration compressors. Any upset in the dew point unit directly impacts gas quality, leading to flaring of the entire export volume up to 5.5 MMSCMD. To reduce this flaring, operations responded by ramping down well production. However, this approach led to reduced gas flow to the sweetening unit, which in turn caused the acid gas compressors to enter unload conditions. This not only compromised the efficiency of acid gas handling but also resulted in additional flaring from associated systems, such as GSU regeneration and TEG units, further increasing the total flared volume.

IMPROVE Phase

Key Points in improving phase:

- Generate Ideas for potential solutions & Prioritization
- Failure Mode and Effects Analysis Failure Modes and Effects Analysis (FMEA) executed
- Pilot was run & X was optimized accordingly

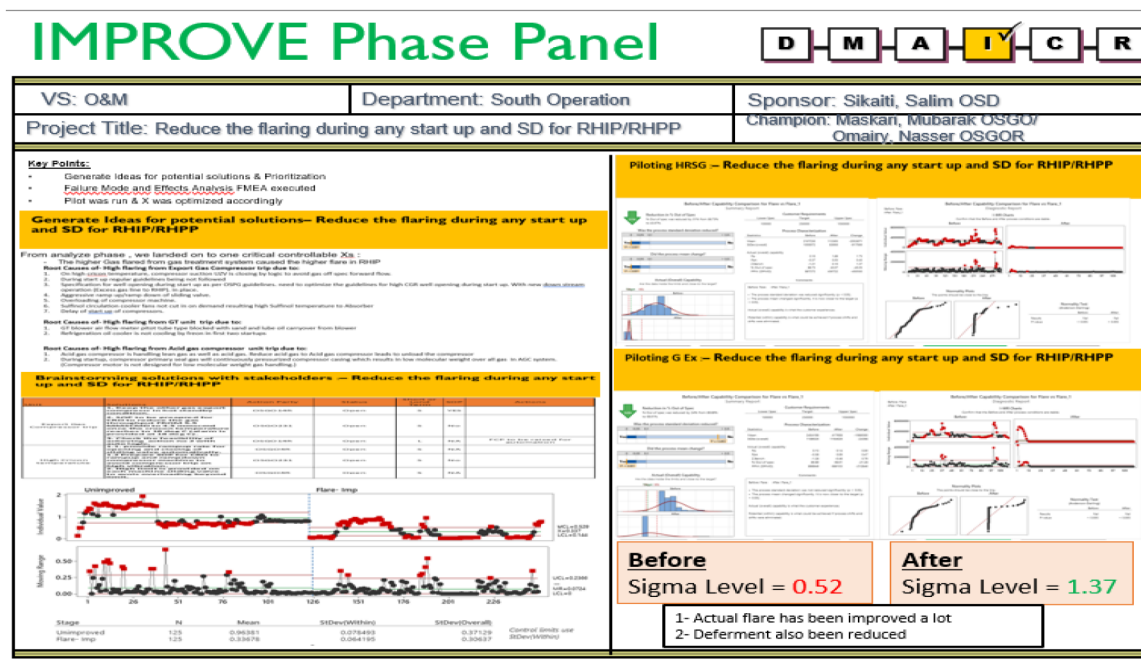


Figure 6—Improve Phase

A pilot was executed in the HRSG system where key control parameters (Xs) were adjusted and optimized based on analysis outcomes. The results showed a significant reduction in actual gas flaring, along with a notable decrease in production deferment. These improvements validated the proposed solutions and provided a solid foundation for full-scale implementation.

The improving phase:

To mitigate flaring and maintain system stability during an HRSG trip, a series of operational steps were established. First, one of the acid gas compressor trains is placed in full recycle mode to prevent it from unloading due to reduced flow. Simultaneously, steam supply and process gas feed to one of the GSU trains are stopped to reduce steam demand and protect gas quality. The load on the remaining HRSG is then increased from 50% to 100% to compensate for the lost steam capacity. Once full steam availability is restored, the previously shut-down GSU is gradually ramped up. As acid gas production increases from the reactivated GSU, the second acid gas compressor train is brought out of recycle mode and placed back into normal load operation, completing the recovery process while minimizing flaring.

One of the key outcomes of this initiative was the development of a detailed Standard Operating Procedure (SOP) outlining quick actions to be taken during startups and unit trips to minimize production deferment and flaring. Given the complexity of operations and the presence of four cockpit areas, this SOP serves as a crucial guide, enabling Control Room Operators (CROs) to respond promptly and accurately based on their respective zones. As illustrated in Figure 4, the SOP provides clear, step-by-step instructions tailored to each cockpit, ensuring consistent and effective decision-making during critical situations figure 7.

5. Development of an action tracker and standing instructions to maintain visibility of assigned tasks and ensure accountability across teams. These measures are designed to embed improvements into daily operations and support long-term sustainability.

REPLICATE Phase

REPLICATE Phase Panel

D

M

A

I

C

R

VS: O&M	Department: South Operation	Sponsor: Sikaiti, Salim OSD	
Project Title: Reduce the flaring during any start up and SD for RHIP/RHPP		Champion: Maskari, Mubarak OSGO/ Omairi, Nasser OSGOR	

Replication plan

Where	Who	When	Comments
RHIP/JK: Production Station	OSGOR/ONOK Team	Q2 2025	TBA by Champion

Follow-up actions:

- SOP is followed and updated
- Ensure monthly Huddles are taking place
- Adherence to CTQ (Control Plan)
- Publish monthly report by the programmer
- LSW is embedded
- Recognition for the Project Team

Figure 9—Replicate Phase

Replication plan

To ensure sustained performance and continuous improvement, the established SOP is actively followed and regularly updated. Monthly huddles are conducted to review progress, share lessons learned, and reinforce team alignment. Strict adherence to the CTQ (Critical to Quality) control plan is maintained, ensuring key process parameters remain within acceptable limits. A monthly report is published by the program owner to track KPIs and document key insights. Leader Standard Work (LSW) is fully embedded into daily routines to support accountability and consistency. Lastly, recognition is given to the project team for their contributions, reinforcing a culture of ownership and excellence.

Conclusion

The implementation of the new operational strategy has led to significant improvements in flare reduction, production stability, and environmental performance. During gas export trips, flaring at R Station was reduced from 928 KSm³ to 213 KSm³ per event, and condensate production deferment dropped from 641 m³ to zero leading to a direct savings of approximately \$200,000 per incident. Additionally, gas export stabilization time improved dramatically, from 18 hours to just 30 minutes.

Similarly, during gas turbine (GT) trips, flaring was reduced from 1,664 KSm³ to 360 KSm³, representing a cost avoidance of roughly \$400,000, while condensate deferment was cut from 1,938 m³ to 360 m³ also worth about \$400,000. These operational enhancements have collectively contributed to an estimated annual reduction of 15,000 tonnes of CO₂ emissions, supporting sustainability goals.

Furthermore, the project identified that a significant portion of annual flaring originates from process upsets in interconnected units, revealing untapped opportunities for additional mitigation. This study underscores the effectiveness of targeted operational enhancements without reliance on major capital investments in achieving substantial flare reduction. Its uniqueness lies in the practical and replicable methodology adopted: enhancing inter-console communication and synchronization across the Utility, Liquid Handling, Gas Processing, and Gas Compression consoles. By focusing on the often-overlooked human and procedural dimensions of transient operations, the approach not only reduced flaring but also

improved startup efficiency and operational responsiveness. Moreover, it established a strong platform for the structured development of Control Room Operators (CROs), presenting a scalable model for other facilities seeking to lower emissions, strengthen cross-functional coordination, and embed sustainable operational excellence.